

Assignment 02: Conceptual Modeling of an Interactive System

System Chosen: Food Delivery Mobile Application (e.g., "FastBite")

Step 1 & 2: Description of the System

Purpose: The application connects users with local restaurants and independent delivery riders. It serves as a centralized platform for browsing menus, placing food orders, processing payments, and tracking delivery status in real-time.

Target Users: Busy professionals, students, and families looking for convenient meal options without cooking or dining out.

User Perspective Workflow: The user opens the app, browses restaurants based on location or cuisine, selects items to add to a virtual cart, completes the checkout process via a digital payment gateway, and then tracks the rider's location on a map until the food arrives.

Step 3: Components and Relationships

Main Components (Entities):

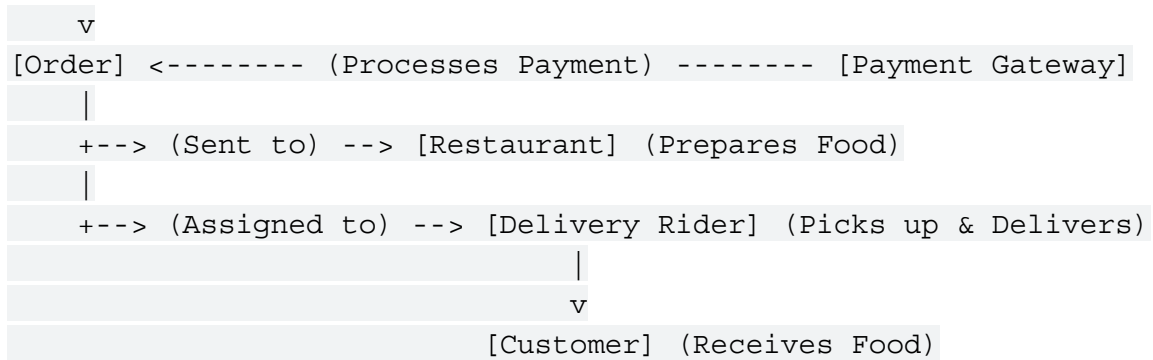
- **People:** Customer (User), Restaurant Manager, Delivery Rider.
- **Information Objects:** User Profile, Digital Menu, Order Details, Payment Record, Real-time Location Data.
- **Physical Objects/Devices:** User's Smartphone, Restaurant's Point-of-Sale (POS) tablet, Rider's Smartphone/Vehicle.

Relationships & Dependencies:

- **Customer ↔ Order:** Customer creates and pays for an Order.
- **Order ↔ Restaurant:** Restaurant receives, prepares, and dispatches the Order.
- **Order ↔ Rider:** Rider accepts the delivery task and updates the Order status.
- **Customer ↔ Rider:** Customer tracks the Rider; Rider contacts the Customer upon arrival.

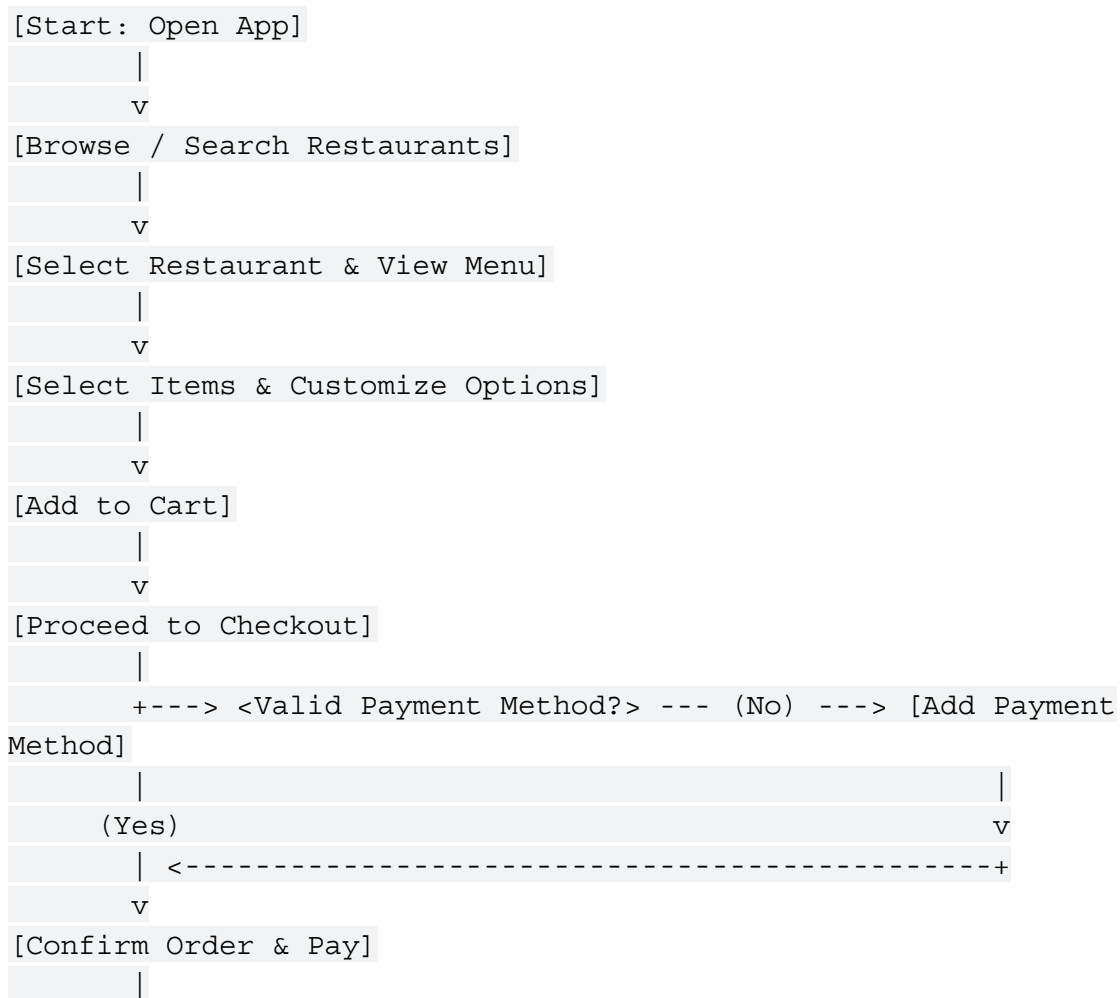
Entity-Relationship Visualization

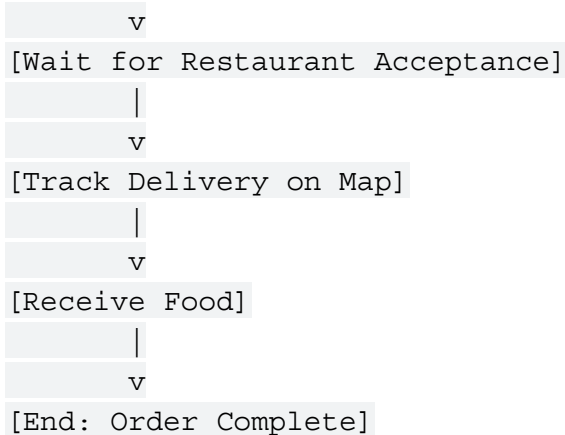
[Customer]
| (Places)



Step 4: Flow Chart (User Task: Ordering Food)

The following flow chart visualizes the user's journey from opening the app to receiving the order.





Step 5: User Functions

1. **Account Management:** Register, login, update address, manage payment methods.
2. **Search and Filter:** Search for specific foods/restaurants; filter by price, rating, or delivery time.
3. **Order Management:** Add/remove items from cart, apply promo codes, submit orders, view order history.
4. **Real-time Tracking:** View order status (preparing, picking up, delivering) and track rider location on a GPS map.
5. **Communication & Feedback:** Call/message the rider, leave reviews and ratings for the restaurant and rider.

Step 6: User Interface and Interaction Mechanism

Interaction Mechanism: Users interact with the system primarily through a graphical user interface (GUI) on a touchscreen mobile device. The interaction is asynchronous during the preparation phase but synchronous during the ordering and tracking phases.

Interface Metaphors:

- *Storefront/Menu:* Digital representation of a physical restaurant menu, categorized by dish types.
- *Shopping Cart:* A persistent icon representing the temporary storage of selected items before purchase.

Input Methods:

- **Touch Gestures:** Tapping (selecting items, confirming buttons), swiping (scrolling through menus, swiping to confirm payment).

- **Text Entry:** Typing search queries, delivery instructions, or discount codes using the on-screen keyboard.

Feedback Mechanisms:

- **Visual Feedback:** Buttons change color when tapped; a loading spinner appears during payment processing.
- **Haptic Feedback:** A subtle vibration when an item is successfully added to the cart or when a payment succeeds.
- **System Status:** Push notifications alert the user at key milestones ("Your order is accepted", "Your rider is arriving").